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SUPERSEDING
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MILITARY SPECIFICATION

LEASHES, DOG, LEAD TYPE: HOLDERS, DOG LEASH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of dog leading leashes and leash holders.

1.2 Classification. Leashes and leash holders shall be of the following types, as specified (see 6.2):

Type I - Leather
Type II - Nylon

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3770

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SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
FF-R-556	- Rivet, Solid, Small; Rivet, Split, Small; Rivet Tubular, Small; Flat Washer (Burr); and Cap, Rivet; General Purpose
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-F-495	- Finish, Chemical, Black, For Copper Alloys
MIL-R-3390	- Rings, Dee
MIL-W-4088	- Webbing, Textile, Woven Nylon
MIL-T-5038	- Tape, Textile and Webbing, Textile, Reinforcing Nylon
MIL-L-11075	- Loops, Strap Fastener
MIL-C-13924	- Coating, Oxide, Black, for Ferrous Metals
MIL-W-27265	- Webbing, Textile, Woven Nylon Impregnated
MIL-W-43061	- Wax, Machine Stitching

STANDARDS

FEDERAL

FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-4-76	- Leashes, Dog Lead Type; Holders, Dog Leash; Type I, Leather; Type II, Nylon Webbing; Assemblies, Details and Sections

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 545 - Steel Wire, Carbon, Cold-Heading Quality, for Machine Screws

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components.

3.2.1 Leather.

* 3.2.1.1 Type I leash and type I leash holder. The leather for fabricating the type I leashes and the type I leash holders shall be inspected in accordance with FED-STD-311. The leashes shall be made from cattlehide leather that has been vegetable tanned or combination tanned with vegetable and alum or vegetable and chrome tanning agents so as to meet the requirements specified herein. The leather shall be furnished in the form of "backs". The color shall be natural as provided by the tanning process. The thickness shall be not less than 10 ounces nor more than 12 ounces when measured as specified in Method 1011. The minimum breaking force shall be 250 pounds when

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tested as specified in Method 2021. The elongation shall be less than 20 percent when measured at a load of 150 pounds when tested as specified in Method 2021.

* 3.2.2 Webbing, nylon. Nylon webbing for the type II leash shall conform to type VIII, class 1 (1 23/32 inch wide) and for the type II leash holder shall conform to type IV, class 1 (3 inch wide) of MIL-W-4088 and shall match Olive Drab 7. The dyed webbing for type II leashes shall be treated to conform to class L of MIL-W-27265.

3.2.3 Tape, nylon. Nylon tape used on the type II leash holder shall conform to type III, 1/2 inch and 3/4 inch wide of MIL-T-5038 and shall match Olive Drab 7.

3.2.4 Thread, polyester. The thread for stitching on leather items shall conform to type I, class 1, subclass C, size No. 3, shade S-1 (C.A. 66022) of V-T-285. The thread for stitching on nylon items shall conform to type I, class 1, subclass A, size F, shade S-1 (C.A. 66022) of V-T-285. The dyed thread shall show fastness to weathering equal to or better than the standard shade sample (see 6.4). If no standard sample is available, the thread shall show good fastness to weathering.

3.2.5 Sealer. For finishing the edges of leather, a colorless aqueous solution of a sealer shall be used. Gum tragacanth has been found to be suitable.

3.2.6 Wax (machine stitching). The wax used as an assist when stitching on leather items shall be golden in color conforming to type II of MIL-W-43061.

3.2.7 Snap, bolt. The round eye swivel snap bolt shall be cast malleable iron, size 7/8 inch and shall conform to North & Judd, No. 5080 or equivalent (see 6.5).

* 3.2.8 Snap, spring. The loop eye spring snap shall be cast malleable iron, size 5/8 inch and shall conform to North & Judd, No. 200 or equivalent (see 6.5).

3.2.9 Loop, strap, fastener. The strap fastener loop shall conform to style B (size 1 1/16 by 1/2), type I or II of MIL-L-11075.

3.2.10 Ring, dee. The dee ring shall conform to configuration X, style 1, size 7/8 by 3/4 of MIL-R-3390.

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* 3.2.11 Rivets, tubular with caps. The tubular rivets shall conform to type XII, class 1, grade E (steel) or F (brass), 9/64-inch nominal diameter of FF-R-556, except that the grade E rivets and caps shall be fabricated from any low-carbon steel between Nos. 1006 and 1012 with wire annealed in process in accordance with ASTM A 545. Caps required for grade F rivets shall conform to type II, grade F cap of FF-R-556. Grade E rivets and caps shall have a black oxide coating conforming to class 1 of MIL-C-13924. Grade F rivets and caps shall have a black chemical finish as specified in MIL-F-495. The length of the rivets shall be such as to provide a secure clinch.

3.3 Construction. Leashes and holders shall be constructed as shown on Drawing 2-4-76 and as specified herein.

3.3.1 Stitching, machine. All stitching shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch. Stitching on type I leash shall be done on a machine using the hot stitching wax specified in 3.2.6. The use of a cold liquid wax or prewaxed thread is not permitted. Thread breaks or skipped stitches on type I leash shall not be permitted. Thread tension shall be maintained so that there is no loose stitching.

3.3.1.1 Repairs to stitching. Defects in type 301 stitching shall be repaired as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching one stitch in back of the end of stitching.

b. Thread breaks, or two or more consecutive skipped stitches or run-offs noted during inspection of the item (in-process or end item inspection) shall be repaired by over-stitching. The stitching shall start one stitch in back of the defective area, continue over the defective area, and continue beyond the defective area on to the existing stitching.

3.3.1.2 Stitching ends (type I items). Thread ends shall be tied and fused.

3.3.1.3 Stitching ends (type II items). The ends of stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem or are held down by other stitching. Box stitching with cross stitch shall have three or more tying, overlapping, or back stitches. Thread ends shall be trimmed to 1/2 inch or less.

3.3.2 Lubrication. The addition of any lubricant to the polyester thread, used with nylon items, prior to or during the sewing operation is prohibited.

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3.3.3 Finishing. All cut edges of leather shall be finished by burnishing with the sealer specified in 3.2.5.

3.4 Marking. The letters "US", manufacturer's name, and the year manufactured shall be indented on leather items and stamped on nylon items in the size characters and in the locations shown on Drawing 2-4-76.

3.5 Workmanship. Leather shall be clean and free of unspecified holes, cuts, tears, or defects such as brand marks, grub holes, scar tissue, hip marks, brittle or bony areas, abrasions, scratches, flanky, spongy or stretchy leather, or fat wrinkles. Webbing and tape shall be free of holes, cuts, or tears and shall not have frayed or scalloped edges. Hardware shall not be broken or malformed and shall be free of any rough edges or corroded areas. Care shall be taken to see that no run-offs, open seams, or needle chews occur.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, the leash or leash holder, as applicable, shall be examined for defects listed in table I and dimensions specified. Any visual defect or any dimension not within the specified requirements shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. The contractor shall furnish a certificate of compliance indicating that the leather is cattlehide back that has been tanned in accordance with 3.2.1.1. In addition, the leather shall be sampled and inspected in accordance with FED-STD-311 to determine compliance with the requirements for thickness, breaking force, and elongation as specified in 3.2.1.1. The lot shall be rejected if one or more specimens fail for any of the above leather requirements.

4.4.2 Process examination. Examination shall be made to determine that hot wax stitching is being performed where specified in compliance with 3.3.1 and that no lubricant is being added to the thread used for the stitching of nylon items for compliance with 3.3.2. Whenever nonconformance is noted, correction shall be made to the item affected and to the process.

4.4.3 End item inspection. A lot shall consist of all completely fabricated leashes of one type or leash holders of one type offered for inspection at one time. The sample unit for this examination shall be one completely fabricated end item.

* 4.4.3.1 Visual examination. Examination of an end item shall be in accordance with the classification of defects in table I. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor) combined defects.

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TABLE I. Classification of visual defects

Examine	Defect	Classification	
		Major	Minor
Leather	Any hole, except where specified	101	
Quality	Any cut or rupture	102	
	Brand mark, healed grub holes, scar tissue, hip mark, brittle or bony area	103	
	Abraded, scratched, or other grain damage		201
	Flanky, spongy, or stretchy leather	104	
	Fat wrinkles		202
Color	Not as specified		203
Webbing, tape	Edge frayed or scalloped		204
Cleanness	Grease or oil stains		205
Hardware	Any component broken or malformed	105	
	Protected finish omitted	106	
	Corroded area		206
	Burrs or sharp edges		207
Snaphook	Spring portion of snaphook loose or defective, causing spring to jam	107	
	Body of snap fails to rotate freely	108	
Construction and workmanship	Any component not fabricated of specified material	109	
	Any rivet not securely clinched	110	
	Any rivet set off center line of body by more than 1/8 inch	111	
	Rivet not capped or capped end of rivet not on side of body specified on Drawing 2-4-76	112	

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TABLE I. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship - continued	Leather edges not burnished		208
	Burnished edges are rough (i.e., not smooth)		209
	Burrs or sharp edges		210
Stitching	Not specified stitch type	113	
	Any open seam not repaired	114	
	Repair of open seam not as specified		211
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped stitches or runoffs occur.		
	Loose stitching resulting in locks exposed on surface of finished item	115	
	Tight stitch tension (i.e., stitches rupture leather, cause puckering on webbing, or may break in use of the end item)	116	
Stitches per inch	One stitch less than minimum specified		212
	Two or more stitches less than minimum specified	117	
	Two or more stitches in excess of maximum specified	118	
	Stitching margin not as specified		213
On leather items	Thread ends not tied		214
	Thread ends not fused		215

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TABLE I. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Stitching - continued			
On nylon items	Thread ends not trimmed to 1/2 inch or less in length		216
Marking	Omitted, incomplete, incorrect, not legible, not of specified size, not in proper location, or not accomplished as specified	119	

* 4.4.3.2 End item dimensional examination. Examination shall be made of the end item to determine compliance with the dimensions specified on Drawing 2-4-76. Any dimensions not within the specified tolerance shall be classified as a defect. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per bundle or per shipping container is more or less than required 1/

1/ Number of bundles to be examined per shipping container (2 bundles).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Each type I and type II leash shall be folded in half twice, and either secured with cotton tape or twine or placed in a commercial 0.001-inch thick polyethylene bag and the bag secured. Five type I holders shall be stacked with dee rings doubled back and alternated end for end and the bundle secured as specified for the leash. Each type II holder shall be placed in a commercial polyethylene bag and the bag closed.

5.1.1.2 Intermediate packing. Leashes or holders of one type only, unit packed as specified in 5.1.1.1, shall be packed in a snug-fitting intermediate box conforming to style RSC, W6c of PPP-B-636. The weight of contents shall not exceed 20 pounds, and the box shall be closed by using method I of PPP-B-636.

* 5.1.2 Commercial. Leashes and holders shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing. Leashes or holders of one type, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, V3c of PPP-B-636 or overseas type style I of PPP-B-601. The gross weight shall not exceed 40 pounds for PPP-B-636 boxes and 200 pounds for PPP-B-601 boxes. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable.

5.2.2 Level B packing. Leashes or holders of one type, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, W5c, or V3s, or V4s of PPP-B-636. Each shipping container shall be closed by using method II of the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Leashes and holders, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate packs, and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

* 6.1 Intended use. Leashes are used for training and handling of dogs. The leash holders are designed to accommodate attachment of leashes and other dog handling equipment at the belt of training personnel.

6.2 Ordering data. Acquisition document should specify the following:

a. Title, number, and date of this document.
b. Type leash or leash holder required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to a standard shade sample of thread, address the contracting activity issuing the invitation for bids.

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6.5 Snaps. No. 5080 bolt snaps and No. 200 spring snaps are available from North & Judd, 699 Middle Street, Middletown, CT 06457. "Or equal" products considered for this use must have the prior approval of the contracting activity.

6.6 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 84
Navy - MC
DLA - CS

Preparing activity:

Army - GL

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